

Crystal structure

half Heusler XYZ
cubic $C1_b$, VEC 18

Machine learning

CatBoost, 45 descriptors
vs DFT: 21.9% median
(cluster held out)

κ_L^{BTE}

Transfer function

$$\kappa_L^{\text{exp}} = \kappa_L^{\text{BTE}} \times \min[c (T/300)^p, 1]$$

$c = 0.42, p = 0.70$

κ_L^{exp}

predicted κ_L ($\text{W m}^{-1} \text{K}^{-1}$)

